

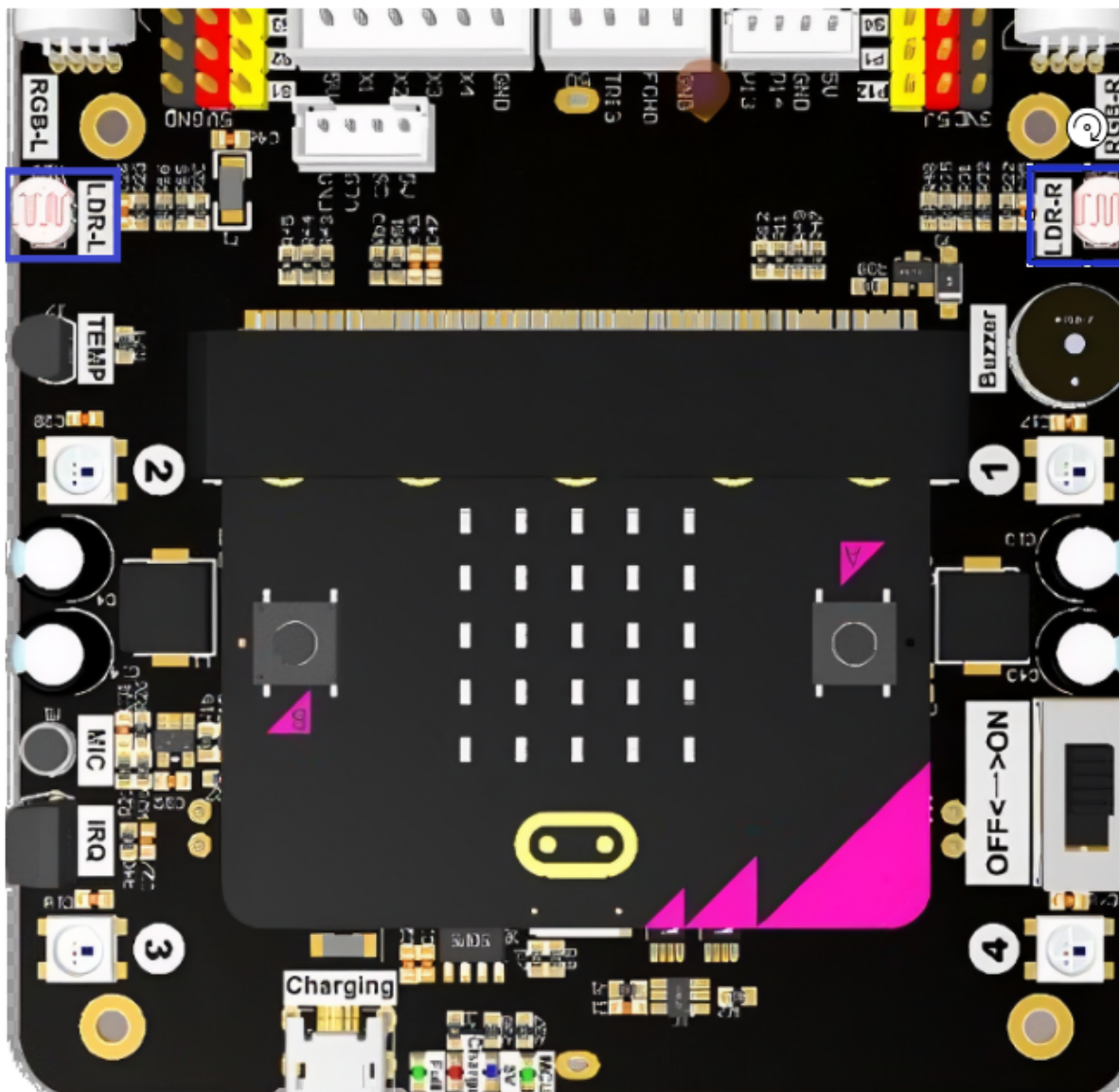
# Photosensitive Sensor Data Collection

## Photosensitive Sensor Data Collection

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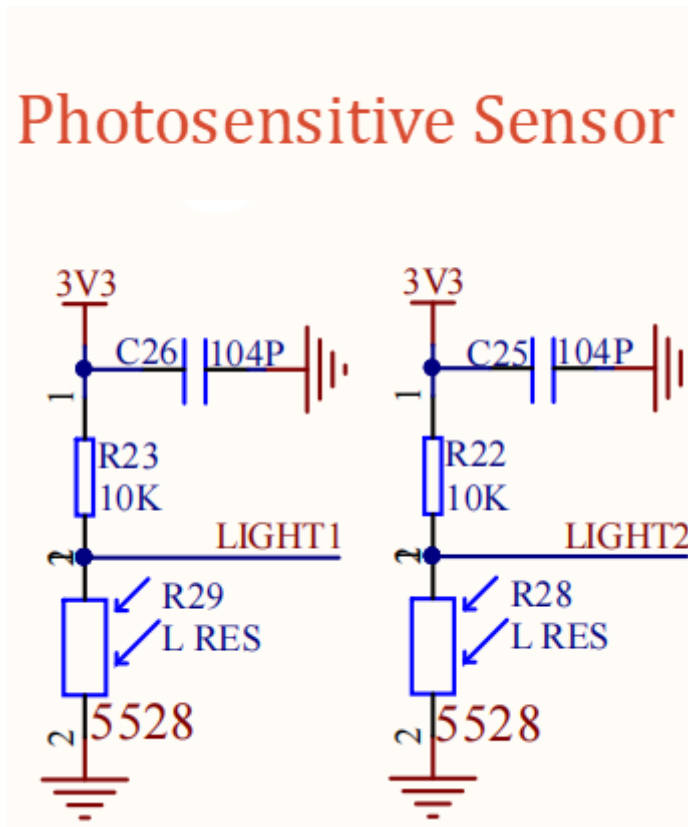
## Experimental Purpose

The Lastbit car is equipped with two light-sensitive (photoresistive) sensors, one on each side, namely LDR-L and LDR-R. In this section, we will learn how to read the data from the light sensors.



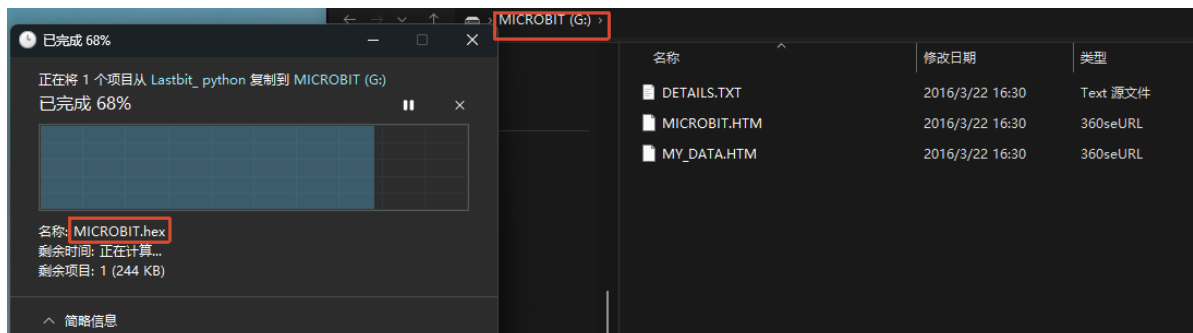
The commonly used material for light sensors is cadmium sulfide (CdS), and its working principle is based on the internal photoelectric effect. The stronger the light, the lower the resistance; when there is no light, it is in a high-resistance state. When the light is dim, the resistance can generally reach 1.5 M $\Omega$ , and when the light is bright, it can be as low as below 1 K $\Omega$ . The light sensor uses a

voltage-dividing light detection circuit to convert the intensity of the ambient light into an electrical signal that the controller can read.



## Experimental Steps

Before flashing, make sure `LASTBIT.hex` has been flashed; otherwise, an error will occur and the module cannot be imported.



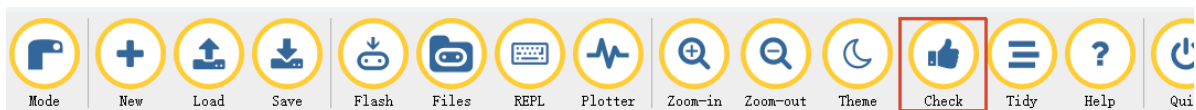
1. Before flashing, switch the switch to the OFF position and connect the computer to the microbit main board using a microUSB cable. **Note: it should be the Microbit interface, not the expansion board's Charging port.**
2. Open the Mu software. Here you can directly copy the program source code we provide; the operation steps are as follows. You can also enter the code yourself in the editor window. Note! All English characters and symbols should be entered in English input mode. Use the Tab key for indentation, and end the last line with a blank line.  
Click the `New` button in the toolbar at the top left, and a file titled untitled will appear.



3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a small red dot after the title bar, indicating that the code content has been modified but is not yet saved.

**Whether or not you save does not affect the code check and flashing.**

4. At this point, you can click `Check` to check whether the code format has any problems.



5. If you have connected the MICROBIT to the PC in advance following the steps above and have flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently, and the editor will show `Copied code onto micro:bit.` at the bottom, indicating that the flashing is complete. After flashing, the indicator light will no longer blink.



6. Unplug the Microusb data cable and switch the switch to the ON position.

## Code Implementation

```
# 2.Light_Sensor_Data_Collection

from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

while True:

    # Read light sensor data
    light_data = Lastbit.read_light()
```

```
if light_data is not None:

    # show left light sensor value
    display.scroll("Left:"+str(light_data[0]))

    # show right light sensor value
    display.scroll("Right:"+str(light_data[1]))

sleep(100)
```

`read_light()` : reads the data of the left and right light sensors. Returns 0 to 255.

`display.scroll()` : scrolls to display the data.

## Experimental Phenomenon

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The Microbit LED matrix will continuously scroll to display the measured values of the light sensors, such as "Left:120 Right:120". The stronger the ambient light, the smaller the light sensor data; conversely, the darker the ambient light, the larger the light sensor data.

## Precautions

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Make sure the car has sufficient battery and the power switch is in the ON position (switched to ON).